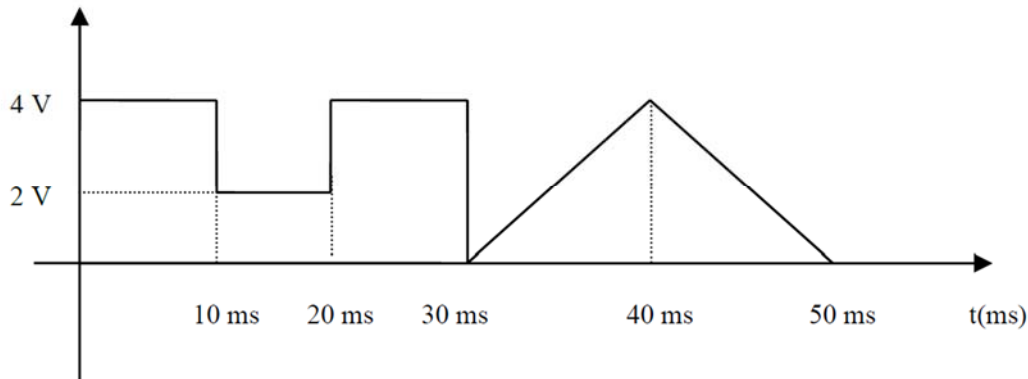


EXERCISES E1

Exercise 1

Given the periodic signal $s(t)$ shown in the figure, with period $T=50$ ms:

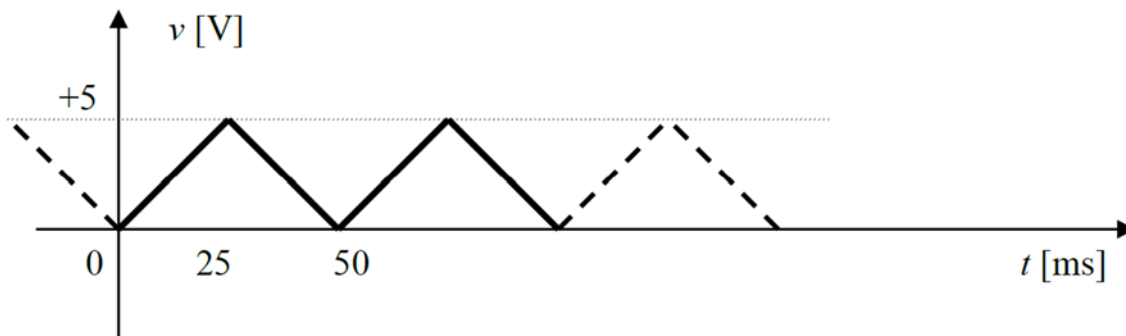


Calculate its Fourier series and signal power.

Let us assume that we have a second periodic signal $w(t)$, with the same period as $s(t)$ and equal to $s(t)$ only between 0 and 30 ms, and constant in the rest of the period. Determine what value $w(t)$ should take between 30 and 50 ms to have the component of order 0 in the Fourier series equal to the analogous Fourier series component of $s(t)$.

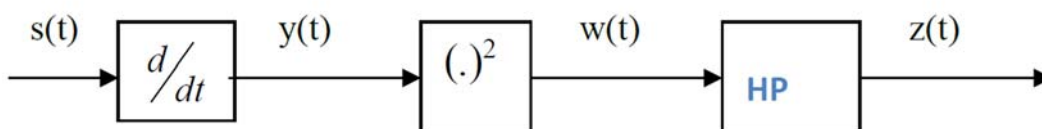
Exercise 2

Given the triangular periodic signal $s(t)$ shown in the graph below, in which the times are expressed in milliseconds and the voltages in volts:



The signal $s(t)$ enters a derivative block with respect to time, and the result $y(t)$ enters a second block that performs the squared elevation, returning $w(t)$.

Finally, the signal $w(t)$ passes through an ideal high-pass (HP) filter, with cut-off frequency $f_T=100$ Hz, which returns $z(t)$.

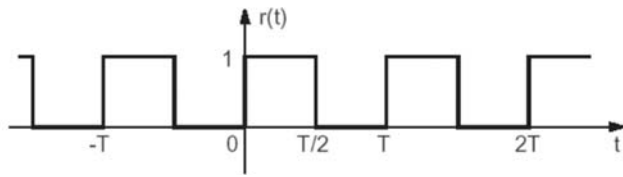


Write the expression of the coefficients c_n of the Fourier series associated with the signals $y(t)$ and $w(t)$.

Plot the waveform output from the high-pass filter and calculate its power.

Exercise 3

Let $r(t)$ be the rectangular, periodic wave function of period T in the figure:



Calculate the coefficients of the Fourier series c_n of its derivative and explain why those of its integral cannot be calculated. Finally, calculate the average value of $r(t)$.

Exercise 4

It is given the signal $s(t) = 2\text{rect}\left(\frac{t-3}{6}\right) \cdot 5\text{rect}\left(\frac{t-4}{2}\right) \cdot 3\text{triang}\left(\frac{t-4}{2}\right)$, where the values are expressed in volts and the times in milliseconds. Draw it and determine its Fourier transform and energy.

Replicate infinite times the part of $s(t)$ included in the interval $0 < t < 6$ thus obtaining a periodic signal $r(t)$. What would be the value of the component at the fundamental frequency of the Fourier transform of $r(t)$?

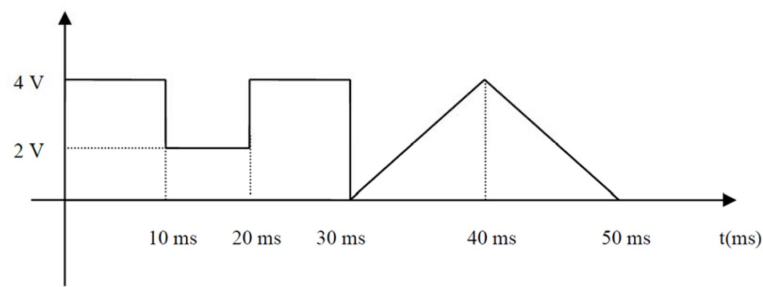
Exercise 5

Suppose we have a sinusoidal signal that is multiplied by another cosine that should have the same period, but actually has period that varies each time and is (each time) equal to k times the period of the first signal, where $k \in [1, 1.1]$ is a random variable with probability density function that is constant in its own defining interval.

The “product” signal passes through an ideal band-pass filter centered at twice the frequency of the sinusoidal signal. The width of the filter is equal to $2/10$ of the frequency of the signal. Express the power of the signal output from the filter as a function of k .

Exercise 1

Given the periodic signal $s(t)$ shown in the figure, with period $T=50$ ms:

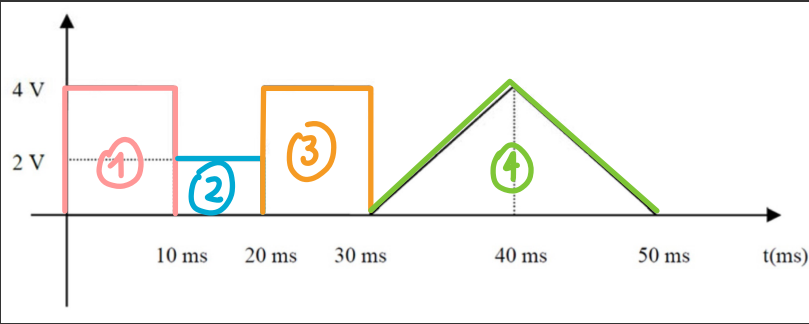


Calculate its Fourier series and signal power.

Let us assume that we have a second periodic signal $w(t)$, with the same period as $s(t)$ and equal to $s(t)$ only between 0 and 30 ms, and constant in the rest of the period. Determine what value $w(t)$ should take between 30 and 50 ms to have the component of order 0 in the Fourier series equal to the analogous Fourier series component of $s(t)$.

PERIOD $T=50$ sec

FOURIER SERIES $s(t) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{2\pi n t}{T_0}}$



$T_0 = 50$ ms

1) FOURIER SERIES

FOR RECTANGLES:

$s(t) = A \text{RECT}\left(\frac{t}{T_0}\right)$
 $c_n = A T \text{SINC}(f t)$
 $f = n/T_0$

FOR TRIANGLES:

$s(t) = A \text{TANGL}\left(\frac{t}{T_0}\right)$
 $c_n = A \frac{t}{2} \text{SINC}^2\left(\frac{f t}{2}\right)$
 $f = n/T_0$

GENERIC RULE:

$c_n = \frac{1}{T_0} s(t) e^{-j 2 \pi f n}$
CENTER OF SIGNAL

1) RECT 1

$c_n^1 = \frac{1}{T_0} A T \text{SINC}(f t) \cdot e^{-j 2 \pi f n}$
 $= \frac{1}{50} \cdot 4V (0ms + 10ms) \text{SINC}\left(\frac{n}{50} 10\right) e^{-j 2 \pi f 5}$
AVERAGE CENTER

2) RECT 2

$c_n^2 = \frac{1}{T_0} A T \text{SINC}(f t) \cdot e^{-j 2 \pi f n}$
 $= \frac{1}{50} \cdot 2V (10 + 20) \text{SINC}\left(\frac{n}{50} 30\right) e^{-j 2 \pi f 15}$

3) RECT 3

$c_n^3 = \frac{1}{T_0} A T \text{SINC}(f t) \cdot e^{-j 2 \pi f n}$
 $= \frac{1}{50} \cdot 4V (20 + 30) \text{SINC}\left(\frac{n}{50} 50\right) e^{-j 2 \pi f 25}$

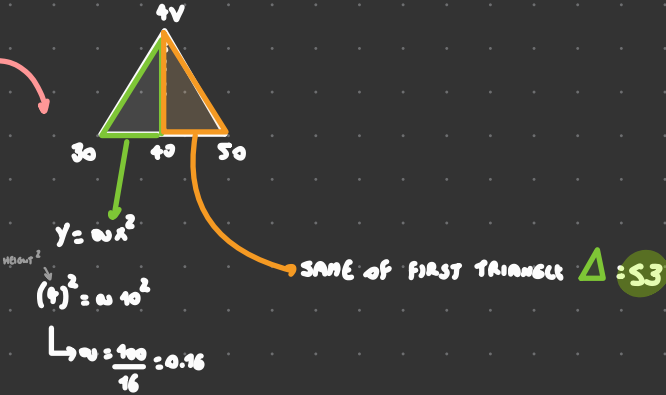
4

$c_n^4 = \frac{1}{T_0} A \frac{t}{2} \text{SINC}^2\left(\frac{f t}{2}\right) e^{-j 2 \pi f n}$
 $= \frac{1}{50} \cdot 4V \left(\frac{30 + 50}{2}\right) \text{SINC}^2\left(\frac{n}{50} 40\right) e^{-j 2 \pi f 40}$

$s(t) = c_n^1 + c_n^2 + c_n^3 + c_n^4$

2) SIGNAL POWER

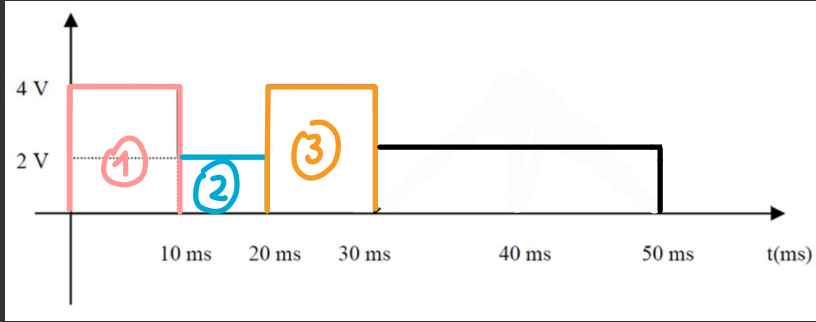
POWER: $P = \frac{1}{T_0} \int s(t)^2 dt = \frac{1}{T_0} (1^2 + 2^2 + 3^2 + 4^2)$
 $= \frac{1}{T_0} (1^2 \cdot 10 + 2^2 \cdot 10 + 4^2 \cdot 10 + 53 \cdot 2)$
COMPUTE "AREAS" $\rightarrow 6 \cdot \frac{1}{2} \cdot h^2$
BASE $\rightarrow$ ALZEDDA ALLA STAMPA Δ



$\int_0^{40} ax^2 dx = a \int_0^{40} x^2 dx = 0.16 \left[\frac{x^3}{3} \right]_0^{40} = 0.16 \left(\frac{40^3}{3} - 0 \right) = 53$

3

Let us assume that we have a second-periodic signal $w(t)$, with the same period as $s(t)$ and equal to $s(t)$ only between 0 and 30 ms, and constant in the rest of the period. Determine what value $w(t)$ should take between 30 and 50 ms to have the component of order 0 in the Fourier series equal to the analogous Fourier series component of $s(t)$.



GENERIC FORMULA

$$C_n = \frac{1}{T_0} \int_0^{T_0} s(t) e^{-j \frac{2\pi n t}{T_0}} dt \longrightarrow C_0 = \frac{1}{T_0} \int_0^{T_0} s(t) e^0 dt = \frac{1}{T_0} \int_0^{T_0} s(t) dt$$

$$C_0^s = C_0^w$$

So find a rectangle (from 30 to 50 ms) with area 33.2

$$\overset{\text{AREA TRAP}}{33.2} = \overset{\text{AREA RECT}}{A^2 \cdot 20} \longrightarrow \overset{\substack{\uparrow \\ \text{HEIGHT}}}{A} \cdot \overset{\substack{\uparrow \\ \text{BASE}}}{20} = 33.2 \longrightarrow A = 2.3$$